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A Method

A.1 VOYAGER Algorithm

Pseudocode 1: VOYAGER algorithm.

```
def voyager(
    environment, # environment that uses code as action space
    curriculum_agent, # curriculum agent for proposing the next task
    action_agent, # action agent for code generation
    critic_agent, # critic agent for self-verification
    skill_manager, # skill manager for adding new skills and skill
    retrieval
):
    agent_state = environment.reset()
    while True:
        exploration_progress = (
            curriculum_agent.get_exploration_progress(
                curriculum_agent.get_completed_tasks(),
                curriculum_agent.get_failed_tasks(),
            )
        )
        task = curriculum_agent.propose_next_task(
            agent_state, exploration_progress
        )
        code = None
        environment_feedback = None
        execution_errors = None
        critique = None
        success = False
        # try at most 4 rounds before moving on to the next task
        for i in range(4):
            skills = skill_manager.retrieve_skills(
                task, environment_feedback
            )
            code = action_agent.generate_code(
                task,
                code,
                environment_feedback,
                execution_errors,
                critique,
                skills,
            )
            (
                agent_state,
                environment_feedback,
                execution_errors,
            ) = environment.step(code)
            success, critique = critic_agent.check_task_success(
                task, agent_state
            )
            if success:
                break
        if success:
            skill_manager.add_skill(code)
            curriculum_agent.add_completed_task(task)
        else:
            curriculum_agent.add_failed_task(task)
```

A.2 Prompting

GPT-4 and GPT-3.5 offer users the ability to designate the role of each prompt message among three options:

- **System:** A high-level instruction that guides the model behavior throughout the conversation. It sets the overall tone and objective for the interaction.
- **User:** A detailed instruction that guides the assistant for the next immediate response.
- **Assistant:** A response message generated the model.

See <https://platform.openai.com/docs/guides/chat/introduction> for more details.

To save token usage, instead of engaging in multi-round conversations, we concatenate a system prompt and a user prompt to obtain each assistant’s response.

A.3 Automatic Curriculum

A.3.1 Components in the Prompt

The input prompt to GPT-4 consists of several components:

- (1) Directives encouraging diverse behaviors and imposing constraints (so that the proposed task is achievable and verifiable): See Sec. A.3.4 for the full prompt;
- (2) The agent’s current state:
 - **Inventory:** A dictionary of items with counts, for example, { ‘cobblestone’: 4, ‘furnace’: 1, ‘stone_pickaxe’: 1, ‘oak_planks’: 7, ‘dirt’: 6, ‘wooden_pickaxe’: 1, ‘crafting_table’: 1, ‘raw_iron’: 4, ‘coal’: 1 };
 - **Equipment:** Armors or weapons equipped by the agents;
 - **Nearby blocks:** A set of block names within a 32-block distance to the agent, for example, ‘dirt’, ‘water’, ‘spruce_planks’, ‘grass_block’, ‘dirt_path’, ‘sugar_cane’, ‘fern’;
 - **Other blocks that are recently seen:** Blocks that are not nearby or in the inventory;
 - **Nearby entities:** A set of entity names within a 32-block distance to the agent, for example, ‘pig’, ‘cat’, ‘villager’, ‘zombie’;
 - **A list of chests that are seen by the agent:** Chests are external containers where the agent can deposit items. If a chest is not opened before, its content is “Unknown”. Otherwise, the items inside each chest are shown to the agent.
 - **Biome:** For example, ‘plains’, ‘flower_forest’, ‘meadow’, ‘river’, ‘beach’, ‘forest’, ‘snowy_slopes’, ‘frozen_peaks’, ‘old_growth_birch_forest’, ‘ocean’, ‘sunflower_plains’, ‘stony_shore’;
 - **Time:** One of ‘sunrise’, ‘day’, ‘noon’, ‘sunset’, ‘night’, ‘midnight’;
 - **Health and hunger bars:** The max value is 20;
 - **Position:** 3D coordinate (x, y, z) of the agent’s position in the Minecraft world;
- (3) Previously completed and failed tasks;
- (4) Additional context: See Sec. A.3.2;
- (5) Chain-of-thought prompting [46] in response: We request GPT-4 to first reason about the current progress and then suggest the next task.

A.3.2 Additional Context

We leverage GPT-3.5 to self-ask questions to provide additional context. Each question is paired with a concept that is used for retrieving the most relevant document from the wiki knowledge base [23]. We feed the document content to GPT-3.5 for self-answering questions. In practice, using a wiki knowledge base is optional since GPT-3.5 already possesses a good understanding of Minecraft game mechanics. However, the external knowledge base becomes advantageous if GPT-3.5 is not pre-trained in that specific domain. See Sec. A.3.4 for the full prompt.

A.3.3 Warm-up Schedule

In practice, we adopt a warm-up schedule to gradually incorporate the agent’s state and the additional context into the prompt based on how many tasks the agent has completed. This ensures that the prompt is exposed to increasing amounts of information over the exploration progress and therefore